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Familial DNA analysis and criminal investigation: Usage, Downsides and Privacy concerns

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Highlights

- Familial DNA analysis is a valuable tool for criminal investigation.
- Strict compliance is necessary for Implementation of familial DNA analysis.
- Ethical, legal and privacy concerns limit the wide usage of familial DNA analysis.
- FDS Necessitates policies developments worldwide for effective usage

Abstract

Since the discovery of Deoxyribonucleic acid (DNA) capability in forensic investigation, it has been an important part of the criminal justice system. In most criminal cases DNA profile originating from evidence sample collected from the crime scene is compared with the DNA profile from the reference sample. However, when a reference sample is not available for comparison, familial DNA analysis can provide important investigation leads in a criminal investigation process by identifying an individual. Moreover, this analysis is also proving effective in the identification of ethnicity and ancestry of an individual. A number of different methodologies and software are being used for familial DNA analysis. This review describes the importance of familial DNA analysis, methodologies used for familial DNA searching and

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identification, and its advantages in forensic. Moreover, ethical, legal and social issues associated with familial DNA analysis have also been discussed along with future directions for the proper implementation of this technology.

Keywords: familial DNA, criminal investigation, ethical issues, genealogy, ancestry

Main text

1. Background

Alec Jeffreys in 1984 used the DNA (Deoxyribonucleic acid) for creating an individual's DNA profile successfully. Since then, DNA has been used for creating a DNA profile from various biological evidence collected from criminal and non-criminal case samples. After the discovery of PCR and use of STR (Short Tandem Repeat) technology, where multiple genetic loci are tested within the genome of an individual, the DNA profiling has become a routine practice in forensic DNA testing laboratories around the world. The DNA profile obtained from the crime scene samples is compared with DNA profile created from the reference sample obtained from the suspect[1, 2]. Generally, DNA profile is created using the CODIS (Combined DNA Index System) autosomal STR loci. These loci have been selected from the highly polymorphic regions in the genome and these selected loci ultimately help in generating unique DNA profile for each individual. Previously 13 STR CODIS loci were used for generating DNA profile, however, in recent years, 20 STR CODIS loci are being used [3, 4]. These loci have been selected from the non-coding regions of the genome, so they do not represent any characteristics related to gender, race or health. In addition, Y-STRs are being used in the forensic DNA analysis, where a male's contributor DNA profile is needed to be separated from the female DNA. This situation is often encountered in the sexual assault cases where evidence sample is often a mixture of male and

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female. Moreover, mitochondrial DNA analysis is also helpful in matching maternal lineages using the sequencing analysis of the hypervariable regions found in the mitochondrial genome [5].

2. Introduction

Familial DNA analysis is the strategy in which biological family members' DNA is used to provide investigative leads for identification of the unknown individual. As one share genetic information very close to those who can be one's close relatives, this information is useful in solving many criminal cases. In recent years, familial DNA analysis has become an emergent field in criminal investigations. Instead of using traditional DNA match searches, familial DNA analysis is used to find a familial relationship [6–8]. It does not take into account the exact matches on the same loci, but instead, similar matches are noted, which can identify the family relationships among the compared individuals. Familial DNA analysis is sometimes left as a last resort in an investigation process when no reference sample can be obtained to compare with the crime scene sample (Fig 1).

Principle lying behind the familial DNA analysis describes that the biologically related individuals, like parents, siblings or closely related relatives will share more alleles on tested loci among themselves as compared to the unrelated individuals. Searching DNA databases for the purpose of familial DNA analysis, 'familial searching' can provide important investigation leads in a case where no comparison could be made with the suspect's DNA due to unavailability of the reference sample [9].

It is very common in the investigation process that a perpetrator's DNA is found on the crime scene but the suspect is not available for comparison, so it becomes difficult to find any

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evidentiary value of the DNA found on the crime scene. In addition, the perpetrator's DNA is also not available in the DNA database for comparison purpose, which makes it difficult to find the real culprit. The real purpose of the familial DNA testing is to search against an available DNA database for the purpose of finding a closely related individual in family relationship to the perpetrator. It has been well established that the more closely related family individuals have more chance of having common alleles at a given locus as compared to the unrelated family individuals [6–8, 10–12].

The use of familial DNA for the forensic searching purpose was first introduced in 2002 in the United Kingdom and it was successfully used for prosecution purpose in 2004. Since its usage in 2002, it has been used successfully in UK for cases where police do not find any investigation lead and they believe that by using familial DNA analysis, they can find a potential investigating clue [7, 10, 13]. Many states of United States have approved the familial DNA analysis for identification of the perpetrator in violent crimes. However, there is a need of well-established policy regarding familial DNA searching in the US[14]. Scientific Working Group on DNA Analysis and Methods (SWGDAM) guidelines have provided information about conducting searches for possible familial DNA analysis, however strict policies are required regarding the protection of privacy of the individual [15] .

DNA databases have provided important investigation leads in solving hideous crimes. However, familial DNA analysis is still a minor part of DNA database searching because of difficulty in interpreting the partial DNA profiles obtained from the database, ethical issues and unclear legal aspects of familial DNA analysis [16]. Instead of simple comparison, familial DNA searching requires additional steps to confirm the familial relationship [7, 10, 11, 15, 17]. When a familial DNA partial match is obtained, it is advisable to conduct Y-haplotype analysis on the

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partial matches obtained from the database [18, 19]. Moreover, it is also helpful to investigate the relatives of the required person to rule out the likely presence of their samples on the crime scene.

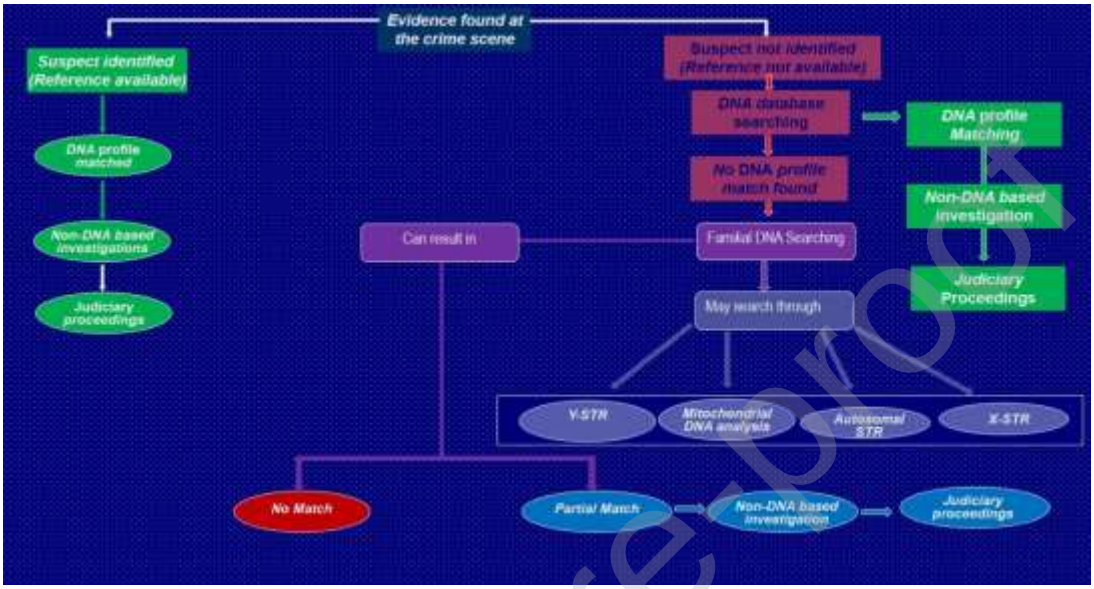


Figure 1. Generalized lay out of familial DNA analysis

3. Methods used in Familial DNA analysis

3.*1. Combined DNA Index System (CODIS) and Familial DNA Searching*

Besides normal DNA testing procedures, familial DNA analysis requires additional stringency for profile matching. DNA profile generated from Genetic Analyzer is added in CODIS in the form of allele numbers and is matched across all the available DNA profiles already available in the software. CODIS software used for the matching of DNA profiles can be used at three levels based upon its stringency, low, moderate and high. The high stringency requires the exact match of the DNA profile, while low and middle stringency generates partial matching results. In criminal investigation procedures, high stringency (exact match) of the DNA profile is considered reliable around the world. However, keeping in view the genetic inheritance of the

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DNA (related individuals show more similarity as compared to unrelated individuals), the low stringency (partial match) capability can also be helpful[7, 17, 20]. Moreover, CODIS is not specifically meant for familial searches, so different software have been designed over the years which lead to effective familial DNA analysis.

In order to counter certain restrictions, various FDS software have been used for familial searching. These software make use of the Likelihood ratio, state or a combination of these two statistical calculations for evaluating familial searching. Additional features of the FDS make it useful for familial DNA analysis as compared to other techniques. FDS software has been used in various countries including, United States (US), United Kingdom (UK) and New Zealand for solving various criminal cases [21, 22]. However, the use of FDS is restricted due to a number of ethical and legal issues raised by scholars and legal bodies. In addition, higher stringency policies, with respect to statistical analysis, are required when evaluating the partial searches obtained from using this software.

3.2 GEDMatch

GEDMatch is another database where autosomal DNA profile is compared with data files of any other DNA testing company to get familial matching. GEDMatch does not offer creation of DNA profile, instead it requires the already generated SNPs based profile to be entered in the database. It gives the output in the form of DNA segments chunks, indicating the matching and non-matching of the query profile with the already available profiles in the database. The database originally founded in 2010, became famous in 2018 with the apprehension of the culprit DeAgnelo. Individuals use this database to find their distant relatives or get information about their parents. A number of different parameters are calculated using this software, such as,

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ethnicity calculator, genetic-distance calculator and many more [23, 24]. Since acquisition of this software by Verogen, inc, the focus of this software has shifted towards solving crimes.

3.3 Ancestry DNA

'Ancestry DNA' is another software used for the familial searching. This software takes samples from the individual who wants to check his/her ancestry. The software makes use of the DNA sequence, in the form of SNPs, against other ancestry DNA in the database and gives the result in the form of 'ethnicity estimator' [25, 26].

3.4 23andMe

23andMe, the first company to give information regarding genetic ancestry to its customers. It is a dedicated personal genomics company which takes saliva sample from the customer, and after DNA analysis (using Single Nucleotide Polymorphism), gives the individual information about the ancestry and health-related information [27, 28] . Approximately three million individuals have been genotyped by this company since 2018.

4. Lineage Testing

In addition to the partial match searches using the above mentioned software, it is effective to use autosomal STR, Y-STR and mitochondrial STR for more reliable results. Y-STR are being employed to explore the paternal lineages of the said evidence sample, while mitochondrial STR analysis helps in exploring the maternal lineage of the evidence sample DNA. These lineage testing, together with the partial matches obtained from different databases, can provide affirmative clue regarding the sample origin which further helps in the identification of the perpetrator [7].

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4.1 Autosomal DNA analysis

Autosomal DNA analysis makes use of the autosomal chromosomes for identifying an individual. These loci have been selected on the ability that they show variability among individuals. The loci are located on those regions of chromosomes which are highly polymorphic in the population. Moreover, these loci have been selected from the non-coding regions of the DNA, therefore, these are not based upon any protein-related qualities. These loci are located on different chromosomes where they can be used for checking individuality of one person as compared to another. Allele frequencies of these loci have been well documented for several population of the worlds and by using these allelic frequencies, rarity of a DNA profile can be assessed using statistical analysis[29–31]. In addition to the autosomal loci, Amelogenin (not STR) is being used for determining whether the DNA profile originated from the male or the female. This analysis has been widely used for criminal and non-criminal cases, for identifying perpetrator, in paternity/maternity cases, in dead body identification and several other cases. These loci are simultaneously amplified in a PCR reaction and fragments are resolved using capillary electrophoresis. The fragments are identified on the basis of their size and fluorescence (primers used in this multiplexing PCR are fluorescently labeled primers), which are then designated as alleles using software and DNA profile is generated.

Familial DNA analysis is generally initiated through the autosomal DNA profile matching. The DNA matching obtained through this process is usually partially matched. However, when familial DNA analysis is required, autosomal loci can only narrow down the individuals [20, 32]. Further confirmation of this partially matched profile is required through Y-chromosome and mitochondrial DNA analysis.

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4.2 Y-STR typing

Prediction of paternal lineage using Y-STR has aided the process of investigation where no reference sample is available for comparison purposes. Y-STRs have been selected from the Y-chromosome and because Y-chromosome is only present in the male and inherited from the father, its haplotype analysis gives crucial information in identification of criminal and determining ancestry. During forensic DNA analysis, Y-STR analysis has gain importance for criminal cases. In this process, male DNA profile can be used effectively for the exclusion purposes[33] In addition, Y-STR analysis is also helpful when there is very little amount of male DNA as compared to female DNA (most often encountered in a rape case). However, familial DNA searching using Y-STR is proving effective in providing investigation leads when unknown perpetrator is involved [34, 35].

4.3 Mitochondrial DNA Typing

In familial DNA analysis, maternal DNA linage determination can provide crucial information during an investigation process. Mitochondrial DNA analysis is very helpful in dead body identification cases because copies of mitochondrial genome are much more as compared to the nuclear genome. Due to higher copy number genome, it is considered effective when nuclear genome is degrades or not available. As mitochondrial DNA is inherited from mother only, mitochondrial DNA analysis can be helpful in narrowing down the potential suspects. This type of DNA analysis is particularly important in analysis of cold cases where evidence is old and biological material is in the degraded form. Mutation rate in mitochondrial DNA has been found ten times greater than the nuclear DNA which makes it extremely useful in determination of the matrilineal changes across the generations. [36]. During mitochondrial DNA analysis,

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sequencing analysis is performed on the Hypervariable regions (HVR-1 and HVR-2) present in the mitochondrial genome. HVR-1 and HVR-2 sequencing is normally performed, however, when used in combination with partial matches obtained from autosomal STR profiling, the results provide reliable information about the perpetrator. Moreover, mitochondrial DNA matching does not individualize a person, rather it establishes the maternal linkage between the compared samples so it should be interpreted correctly otherwise DNA evidence weight may get diminished in the court [22, 37, 38].

4.4 Single Nucleotide Polymorphism (SNP) genotyping

In most of the commercially available databases for familial DNA analysis, SNP genotyping is used. '23andMe', a software used for genealogy analysis and familial DNA testing, makes use of the SNP genotyping when predicting a result. Single Nucleotide Polymorphism analysis confers several important advantages over the Short Tandem Repeat analysis. As they show low mutation rate and are very abundant in the human genome, they can offer an effective way of analyzing DNA. Moreover, due to their nature of analysis, these are extremely useful in degraded DNA samples where long fragments of DNA are not likely to be found. However, a large number of SNPs are tested in the human genome for the proper identification of the individual. During this testing, genetic variations which are more common in the individuals of a species, are used for the identification purposes. Currently many DNA testing laboratories are using SNP for identification and familial DNA analysis [39–41].

5. Potential usage of familial DNA

Familial DNA analysis has gained tremendous importance in recent years. In the absence of any standard reference sample, familial DNA analysis provides important investigation clues [42,

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43]. This situation is encountered in a large number of cases during investigation proceedings Partial DNA matches obtained from the crime scene sample with the available database samples has led to many important investigation leads, which ultimately resulted in the identification of the culprit in those cases.

The importance of the familial DNA searching can be well illustrated by the Joseph James DeAngelo, claimed as the 'Golden State Killer'. During two decades, 1970s and 1980s, the DeAngelo had committed a number of murders and more than fifty sexual assault cases. His profile was found on different shreds of evidences collected from the crime scene, but it was difficult to identify him due to the non-availability of suspect's DNA. However, after a long span of forty years, a breakthrough appeared when an officer in the California ran a genealogical search database GEDMatch for the DNA profile obtained from the evidence samples. The profile match was found to be of a third cousin of DeAngelo. By tracing through family tree, DeAngelo was narrowed down. His sample was taken as a reference and when the DNA profile was generated, it matched with the evidence sample DNA profiles obtained in the past [44–46]. The investigation led to the apprehension of the criminal and this case led to the expansion of various ancestry based databases worldwide.

Familial DNA analysis is not only helpful in the identification of the criminal, but also for the identification of the deceased in case of mass natural disaster victim's identification or in other cases, such as plane crash etc. Those cases where reference samples of the offspring or parents are not available, use of familial DNA analysis can help in the identification of the individuals [7, 47]. However, familial DNA analysis in case of dead body identification should be performed cautiously and requires an additional statistical tool for verification, as normal statistical analysis such as Likelihood Ratio (LR), may lead to false identification process.

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6. Downsides of familial DNA analysis

There are a number of issues concerning the familial DNA analysis. Among many, usage of familial DNA analysis can lead to false hit. As this process is based upon the partial matches obtained by comparing two DNA profiles, this can lead to individuals who have nothing to do with a crime. In most of the countries, no uniform policy persists regarding the usage of familial DNA analysis [14, 22].

Another important point of discussion is 'security'. In the absence of proper policies for entering and analyzing DNA profile in the different databases available for the comparison purpose, the DNA profile is not secured. The usage of the familial DNA searching has increased the risk of dissemination of the DNA profile information of a person. Moreover, a given DNA profile is also matched across different DNA databases for a potential match; this can also hamper the security of the individual's DNA profile [48, 49].

Statistical analysis is another crucial factor when dealing with familial DNA matching. The analysis of the partial matches obtained from DNA comparison during familial searches calls for additional scrutiny when concluding a 'hit'. It has been observed that certain statistical calculation methods, such as, Likelihood Ratio (LR), Random Match Probability (RMP) should be applied with some correction factors. If no modifications are made on the above mentioned statistical calculation methods, this can lead to misinterpretation and wrongful weight of familial DNA evidence [47, 50]

When searching for familial DNA in solving a crime, care should be made at the investigation stage. Partial DNA matches, when not interpreted properly, may lead to an individual who was never there on the crime scene. In the case of the criminal investigation

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process, familial DNA searching can be accurately known as a ‘narrowing down technique’ rather than a ‘match technique’. However familial DNA analysis when coupled with the crime scene investigation, it can lead to a more reliable hit [10, 51].

7. Ethical and Legal Issues with familial DNA analysis

With the increasing application of familial DNA analysis, there are a number of ethical issues. As familial DNA searching is based upon the partial matches and probability estimations, this may increase the number of individuals for investigation [52]. Moreover, there many database searching for familial DNA analysis are racially biased, containing more data from one color of individuals as opposed to equal DNA profiling. In most criminal investigation databases, DNA profiles from the criminal offenders are more as compared to the non-criminal individual. This can lead to a biased partial hit [53]. In the recent years, the development of the databases for familial searches are based upon the inequality, that is, people belonging to one ethnicity or convicted offenders, have more chances that their relatives are included in the investigation as compared to others. This may cause wrongful identification of the suspect and ultimately led to the wrongful conviction of the individual [54].

Keeping in mind the family structures and how they manage their relationship, familial DNA analysis accounts for major legal and ethical issues. The DNA profile of a person which is once entered in the database raises major concerns as this profile may have an impact on the family and relatives of the person. Moreover, it may interfere with the human right of the person because ‘not to know’ about whatever is happening with his DNA profile in the database and impact on his/her family [14, 24, 55]. In addition, there are several genetic loci which have been identified to be associated with genetic diseases. Probabilistic chances of inheriting such diseases

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form the parent and transferring on to the offspring has far-reaching effect on the life a person Hence, DNA profile in the DNA database raises several implications to consider for a person's basic right.

8. Future perspectives

Application of the familial DNA analysis in the criminal investigation has proved effective in recent years. This analysis requires more stringent conditions as compared to other forensic DNA analysis. Use of probability calculating software, with added correction factors according to the ethnicity in question, can improve the searching capability of familial DNA analysis. Partial matches obtained by DNA comparison can be coupled with the mitochondrial analysis, Y-STR analysis and by use of SNPs. Adopting multiple comparison DNA techniques can solve two purposes in future; it can reduce the chance of error and, it can help minimize many ethical, legal and social issues associated with familial DNA analysis.

9. Conclusion

Use of familial DNA analysis in forensic science has added a new dimension in the criminal investigation system. It has been helping law enforcement agencies in finding an individual where there is no reference sample is available for comparison purpose. A number of government and private founded laboratories and databases are available for readily making comparison among profiles. However, the process should be conducted in stringent conditions. As partial matching results are obtained from DNA profile comparison, misinterpretation of the results can lead to wrongful convictions. In addition, several ethical, social and legal issues still

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persist when familial DNA analysis is used for criminal investigation proposes which asks for uniform policy making and legalization of this process.

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Consent for Publication

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Conflict of interest

The authors declare that there are no conflict of interest.

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